

In a Different Voice: Gender Composition and Scientific Writing in Economics*

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Abstract

How scientific ideas are communicated can shape how they are evaluated, disseminated, and ultimately integrated into cumulative knowledge. This paper studies whether author-team gender composition is associated with differences in scientific writing in economics. Using 9,117 abstracts from top general-interest economics journals and a 16-item LLM-based rubric, we first benchmark the LLM readability measure against Hengel’s established readability metrics. We then examine mean differences across writing dimensions and estimate regressions with extensive journal, year, article, and author controls. Female-majority teams write more readable, less jargon-heavy, and more direct abstracts, and they generally provide more evidentiary support. They also make weaker novelty claims. By contrast, differences in hedging, qualifiers, assertiveness, and emotional valence are small or inconsistent.

Keywords: Gender differences; scientific writing; readability; natural language processing; team composition.

JEL Classification: J16; C55; C81.

*This paper is a work in progress intended solely for feedback and should not be circulated.

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1 Introduction

Scientific writing is not a neutral container for ideas. The way results are framed, in particular, how directly arguments are stated, how much jargon is used, whether evidence is foregrounded, and how strongly novelty is claimed, can shape both peer evaluation and the dissemination of knowledge. This possibility echoes a broader concern, associated with the idea of a “different voice” (Gilligan 1993), that dominant evaluative standards may overlook systematically different modes of communication. Yet empirical work in economics has focused much more on gender differences in publication, promotion, and review outcomes than on the writing itself.

This paper asks whether the gender composition of author teams is associated with systematic differences in how economists write. The paper builds most directly on Hengel 2022, who shows that female-authored papers are more readable and appear to face higher writing standards in peer review. We extend that insight beyond readability by measuring a broader set of surface-level writing features in abstracts, including directness, jargon, evidence and citation usage, active voice, novelty claims, hedging, and related rhetorical dimensions. The objective is not to infer author intent or personality from text. Rather, it is to identify whether different teams systematically present research in different ways.

Our data come from Hengel 2022, with the main analysis conducted on 9,117 abstracts from four top general-interest economics journals between 1950 and 2015. Each abstract is evaluated using a 16-item LLM-based rubric covering readability, technicality, directness, evidentiary support, and rhetorical style. Because LLM-based measures may invite skepticism, the empirical analysis proceeds in three steps. First, we benchmark the LLM readability score against the readability measures used in Hengel 2022. Second, we examine unconditional mean differences across all 16 criteria. Third, we estimate regressions that add journal, year, article, and author controls to assess whether the observed patterns persist after accounting for observable compositional differences.

The validation exercise shows that the LLM readability measure tracks the same underlying pattern documented by Hengel 2022. In unconditional comparisons, female-majority teams score higher on readability, directness, accessibility of language, and ev-

idence/citation usage, and they make weaker novelty claims. Once controls are added, the clearest differences remain higher readability, less jargon, greater directness, more evidentiary support, and weaker novelty claims among female-majority teams. By contrast, differences in assertiveness, qualifiers, emotional valence, and several other tonal dimensions are small or unstable across specifications. The results therefore do not support a simple story in which women write in a uniformly more hesitant or emotional way. The clearest differences are concentrated instead in clarity, technical density, and the organization of claims.

This paper contributes to three literatures. First, it adds to work on gender and evaluation in economics by studying writing style as a potential input into evaluation rather than focusing only on publication or career outcomes. Second, it extends readability analysis from Hengel 2022 to a multidimensional measurement framework that separates clarity and structure from tone and affect. Third, by showing which dimensions move together with team gender composition and which do not, it sharpens the debate over what a “different voice” in economics actually consists of.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data, authorship measures, and LLM rubric. Section 4 outlines the empirical strategy. Section 5 presents the validation exercise, mean-difference analysis, and controlled regressions. Section 6 concludes.

2 Related Literature

This paper sits at the intersection of two literatures.¹ The first concerns gender, voice, and communication. Gilligan 1993 provides the motivating intuition that dominant evaluative frameworks may overlook alternative ways of presenting judgment and reasoning. Although that literature is not about economics articles specifically, it suggests that apparently neutral standards of professional communication may embed preferences for particular styles of exposition.

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The second is the economics literature on gender and the production and evaluation of research. Card et al. 2020 study whether referees and editors in economics treat submissions differently by gender, and Sarsons 2017 shows that collaborative work can be differentially recognized by gender. Most directly, Hengel 2022 documents that female-authored economics papers are more readable and that the writing gap widens during peer review, consistent with women facing higher writing standards in publication.

This paper contributes to that literature in three ways. First, rather than studying editorial decisions, citations, or promotion outcomes, we focus on writing itself as an observable input into evaluation. Second, we move beyond a single readability measure by using a multidimensional rubric that distinguishes clarity and structure from tone, assertiveness, and affect. This matters because several common narratives about gendered writing, such as those centered on hedging, confidence, and emotionality, receive limited support in our data. Third, by organizing the evidence in three steps, validation of the LLM measure, unconditional mean differences, and controlled regressions, we show both that the measurement is anchored to existing work and that the main patterns are not driven solely by compositional differences across journals, years, or paper types.

3 Data and Measurement

The data used for this paper are drawn from the publicly available replication package for Hengel 2022, hosted on GitHub. The data package covers 11695 English-language articles from the years 1950 to 2015 from 5 top journals in economics: American Economic Review (AER), Quarterly Journal of Economics (QJE), Econometrica (ECMA), Review of Economic Studies (ReSTUD), and Journal of Political Economy (JPE). The majority of the analysis in this paper excludes ReSTUD, which Hengel 2022 includes primarily for its availability of submission-acceptance dates, and AER Papers & Proceedings, which do not contain abstracts. A full breakdown of articles by journal and decade is in Appendix B.

The article data includes the title, abstract, citation count, submission and acceptance date, JEL codes, and pre-computed readability scores from Hengel 2022. The article data

is linked with author-level data on gender, native language, and institution.

For this paper, we do not re-estimate readability scores or re-verify the gender coding of Hengel 2022. We also exclude any analysis of full-text readability, as processing the full text through the LLM would be prohibitively expensive at scale and because evaluation quality degrades with longer inputs (Modarressi et al. 2025).

Following Hengel 2022, our main specification classifies papers as 'male-authored' when women constitute a strict minority of the author team. For papers where women are a majority, the gender variable is defined as the ratio of female to male authors.

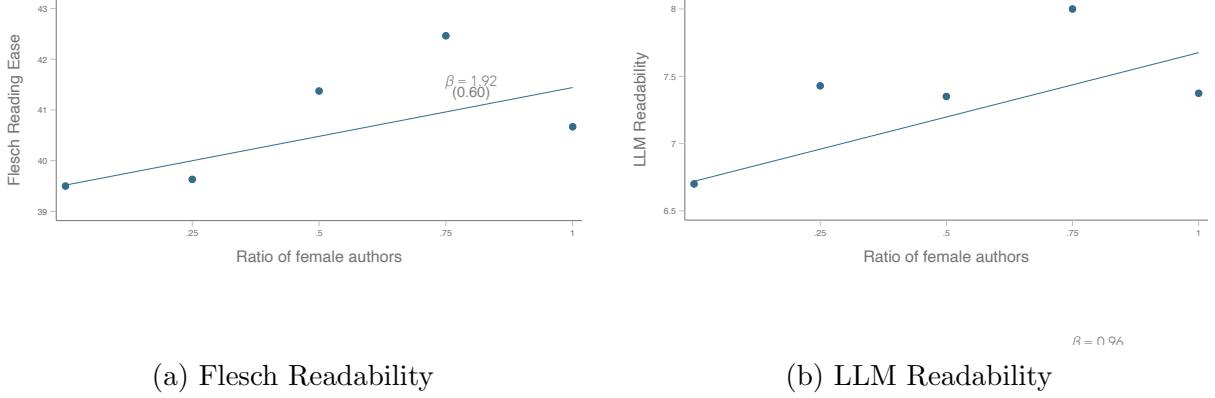


Figure 1: Relationship between readability and the ratio of female authors

Figure 1 plots readability against the female author ratio for two measures: Flesch readability (copied from Hengel 2022) and LLM readability. They both show a broadly positive relationship, they differ in shape. We also estimate our main results under alternative authorship specifications:

- Using a binary equal to 1 if the article’s authors are majority female
- Using a binary equal to 1 if there is 1 female coauthor
- Using only single-gender papers
- Using only solo-authored articles
- Comparing articles with a senior female coauthor to entirely male-authored papers

Further description of the data can be found in Section 2 and Appendices C and D of Hengel [2022](#).

3.1 Tonal Analysis Rubric

Additionally, we derived 16 variables based on the abstract. We use a 16-item rubric to evaluate paper abstracts on the scale of 1-10 for Modal Verb Strength, Hedging Frequency and Type, Qualifier Density, Acknowledgment of Limitations, Caution-Signaling Connectors, Assertiveness and Voice, Active/Passive Voice Ratio, Sentence Length and Directness, Imperative-Form Occurrence, Pronoun Commitment, Novelty-Claim Strength, Jargon and Technicality Density, Emotional Valence, Evidence and Citation Usage, Practical and Impact Orientation, Readability. See exact evaluation rubric in Appendix: [Rubric](#).

3.2 LLM Evaluations

To conduct the tonal evaluations, we developed a Large Language Model (LLM) powered algorithm that would take the abstracts from the papers, process them into a readable LLM message, and then pass the message through to the LLM, which would in turn give us a structured output of evaluations. The LLM model we chose for this analysis was GPT-5, which includes both standard LLM capabilities and the ability for more complex reasoning. We valued the model’s ability to generate detailed thinking processes before answering to handle the complexity of our multi-dimensional tonal analysis rubric, as reasoning models have performed better than non-reasoning models as task complexity increases ([Shojaee et al. 2025](#)). This also allowed us to keep our input token count to a minimum, which increases model accuracy ([Modarressi et al. 2025](#)).

This process of passing the abstracts and rubric to the model are sufficient to generate the needed evaluations, but insufficient in providing the most accurate responses. To increase response accuracy, we implemented prompt engineering, a process of creating clear instructions for the model to increase performance ([OpenAI 2025](#)). Part of this process is to induce step-by-step thinking in the model and allow for model reasoning for each section of the rubric, both shown to increase model accuracy ([Zhang et al. 2022](#)).

To validate the accuracy of the evaluations, we first trained the algorithm on a small subset of abstracts, auditing the evaluation output and reasoning for these responses and using that to improve our prompt. We then tested the algorithm on a different subset of abstracts to validate the output and reasoning before running the algorithm on the full dataset. A separate validation was performed using readability scores, comparing our outputs against Hengel (2022) manually calculated readability scores, checking to see if our evaluations were consistent with hers at the population level.

Despite the validation involved with calibrating the LLM for our project, there is still some randomness involved in the LLM evaluations. Because of this, the standard errors for analysis using LLM evaluations does not fully reflect the instability of our estimates.

4 Empirical Methodology

Our empirical analysis proceeds in three steps. First, we validate the LLM-based readability score by benchmarking it against the readability measures used in Hengel 2022. This shows that the LLM measure is anchored to an existing and interpretable construct rather than being an arbitrary score. Second, we present unconditional mean differences across the 16 standardized writing-style criteria. Third, we estimate regressions that add journal, year, article, and author controls to assess whether the observed differences survive after accounting for observable compositional differences across papers and teams. Before we present our formal analysis, we describe summary statistics of our data.

4.1 Summary Statistics

In total, across the top 4 journals, we have 9117 articles, written by 16055 authors, of whom fewer than 10% are women. Out of all articles, 8264 are single gender out of which only about 3% are female-only articles. About 13% of articles have at least 1 female author, and 9% have a majority of female authors.

Table 3 in Appendix B gives the mean and standard deviations for the LLM evaluations of every rubric item. This table also clearly shows the outlines of a gender writing

difference, with Readability, Sentence Length, and Directness, and Acknowledgment of Limitations showing large differences in means between the all-male and all-female populations. Further summary statistics are available in Appendix B.

4.2 Analysis

Hengel 2022 establishes that female-authored papers are written more clearly than comparable male-authored papers. We replicate and extend this result by substituting first our LLM based readability score to confirm consistence of LLM evaluation on manual readability scores from Hengel 2022 and next remaining 15 LLM-based evaluation metrics for tonal analysis.

To estimate the relationship between author gender and writing characteristics, we regress respective outcome variables on the female authorship measure, controlling for various article- and author-level variables, using different specifications. The estimating equation is:

$$R_j = \beta_0 + \beta_1 \text{female ratio}_j + \gamma \mathbf{X}_j + \delta \mathbf{Y}_j + \varepsilon_j \quad (1)$$

where R_j is the readability score or other LLM based tonal criteria for article j , female ratio_j is the female authorship measure described in Section 3, $\mathbf{X}_j$ is a vector of article-level controls, $\mathbf{Y}_j$ is a vector of author-level controls, and β_1 is the coefficient of interest. To the original analysis, we add our readability score and the rubric’s tonal criteria group composite scores as dependent variables. For a full discussion of identification, see Hengel 2022 Section 3.2.

5 Results

5.1 Benchmarking the LLM Readability Measure

We start by with readability score to make comparison with Hengel 2022. Table 1 shows the results of regressing various readability scores on the ratio of female authors. The first 5 rows of Table 1 replicate the corresponding specifications from Hengel 2022. The

final row adds the LLM readability score as the dependent variable.

Results for Table 1 suggest that abstracts written by women score between 0.27 and 0.4 points higher on average on our rubric than those written by men after accounting for different levels of controls. The estimates are significant at the one percent level for all levels of controls, though this does not fully account for possible inconsistencies or noise in the LLM evaluation process, meaning the true standard errors are likely higher.

Appendix C.1 shows Table 1 estimated using the alternative specifications for female authorship described in Section 3. These alternative specifications yield similar coefficients and statistically significant results for the LLM readability metric, indicating robustness.

5.2 Mean Differences in Writing Style

After establishing consistence of LLM based readability score with manually defining readability scores from Hengel 2022 we look at other tonal criteria. We start by comparing sample means for male and female for each of the 16 rubric items. To define gender of a paper we use gender of majority. That is if at least 50% are female we call it female and male otherwise. To make comparability across rubric items, we standardized all criteria with mean and standard deviation. Figure 2 shows difference in means with 95% intervals for each of the 16 standardized criteria.

Table 1: Gender differences in readability, article-level analysis

	1950–2015					1990–2015			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Flesch	1.25 ** (0.53)	1.23 ** (0.52)	1.23 ** (0.53)	1.26 ** (0.54)	1.49 *** (0.56)	1.31 ** (0.59)	1.36 ** (0.54)	1.29 ** (0.60)	1.94 ** (0.77)
Flesch-Kincaid	0.23* (0.13)	0.22* (0.13)	0.23* (0.13)	0.24* (0.13)	0.26* (0.14)	0.34 ** (0.15)	0.37 *** (0.14)	0.34 ** (0.15)	0.46 *** (0.17)
Gunning Fog	0.36 ** (0.15)	0.36 ** (0.15)	0.37 ** (0.15)	0.40 ** (0.15)	0.42 *** (0.16)	0.51 *** (0.18)	0.50 *** (0.15)	0.51 *** (0.18)	0.60 *** (0.19)
SMOG	0.25 ** (0.10)	0.25 ** (0.10)	0.26 ** (0.10)	0.27 ** (0.11)	0.29 ** (0.11)	0.32 ** (0.12)	0.31 *** (0.11)	0.33 ** (0.13)	0.40 *** (0.14)
Dale-Chall	0.08 (0.05)	0.08 (0.05)	0.08 (0.05)	0.08 (0.05)	0.09 (0.05)	0.11* (0.06)	0.11* (0.06)	0.11* (0.06)	0.17 ** (0.07)
LLM Readability	0.36 *** (0.07)	0.35 *** (0.07)	0.35 *** (0.07)	0.40 *** (0.08)	0.32 *** (0.08)	0.39 *** (0.08)	0.31 *** (0.06)	0.38 *** (0.07)	0.27 *** (0.06)
No. obs.	9.117	9.117	9.117	9.117	9.117	5.211	5.211	5.211	5.774
Editor	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blind	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year		✓							
Journal×Year			✓	✓	✓	✓	✓	✓	✓
N_j				✓	✓	✓	✓	✓	✓
Institution				✓	✓	✓	✓	✓	✓
Quality				✓	✓ ¹	✓ ¹	✓ ¹	✓ ¹	✓ ¹
Native speaker				✓	✓	✓	✓	✓	✓
JEL (primary)							✓	✓	✓
Theory/empirical									
JEL (tertiary)									✓

Notes. Figures represent coefficients on female ratio from 45 separate OLS regressions of readability scores on the ratio of female authors (papers with fewer than 50 percent female authors are classified as male, see the text). (6)–(9) are estimated on the sample of papers published on or after 1990 with a primary JEL code; (9) includes 563 articles from AER P&P (see the paper). Quality controls denoted by ✓¹ include citation count (asinh), max. T fixed effects (author prominence) and max. t (author seniority). Standard errors clustered on editor in parentheses. ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

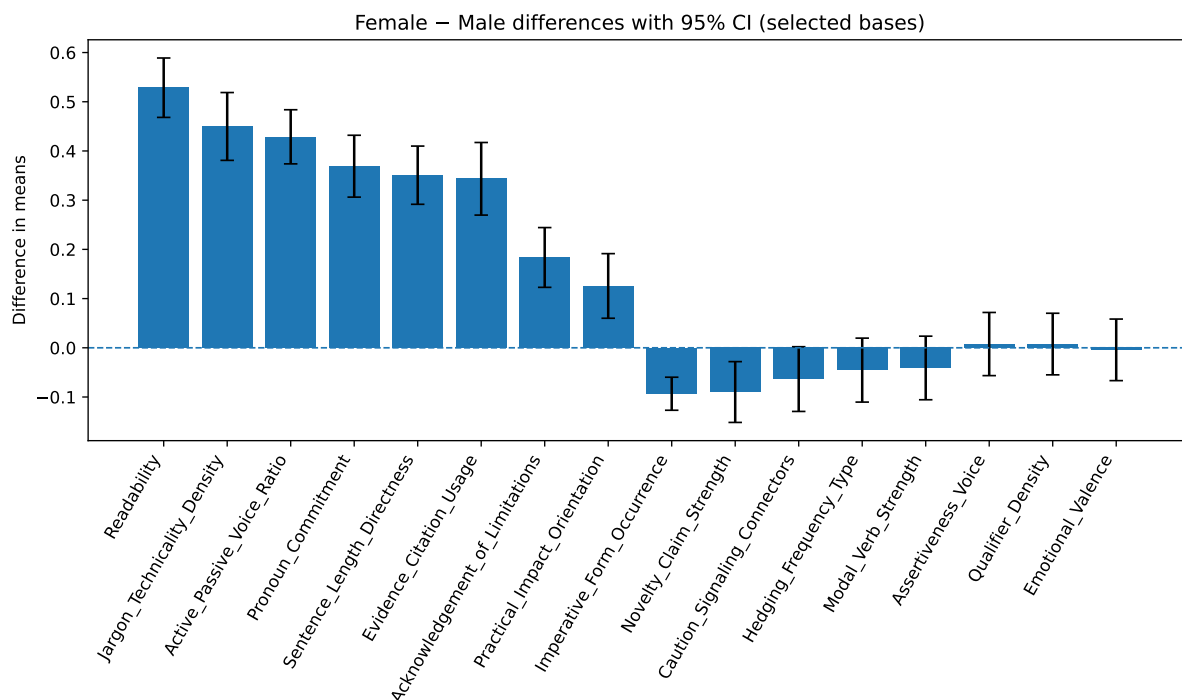


Figure 2: Papers are classified as female if at least 50% of authors are female, and male otherwise. All criteria are standardized (mean zero, unit standard deviation) for comparability. Figure shows differences in means with 95% confidence intervals across the 16 criteria. The sign of “Jargon Technical Density” is reversed so that positive values indicate less jargon for female-majority papers; for all other criteria, positive values indicate higher scores for females.

Taken together, the evidence points to systematic and robust differences in writing by team gender composition. Female-majority teams emphasize clarity – higher readability and greater directness – use more active voice, avoid jargon, and provide more evidentiary support through citations. Differences in acknowledgment of limitations are statistically significant but modest in magnitude; under our coding, positive values indicate that female-majority teams make fewer explicit limitation statements (see Appendix A for the scale). There is also a smaller but significant tendency for female-majority teams to place greater emphasis on practical importance.

In contrast, differences in tone – such as assertiveness, qualifiers, hedging, and emotional valence – are generally small or negligible, with the exception of slightly weaker novelty claims and fewer imperatives among female-majority teams.

5.3 Controlled Regressions.

It is possible that the mean differences are simply due to Female or Male groups working on different type of research. To control for such compounding we control for journal, publication year, and article characteristics.

Tables 11 and 12 in appendix C.2 show results. For the full sample (columns 1 to 5) shows that there are statistically significant positive (female higher) differences in Sentence length, technical jargon, Evidence/citation usage and readability, while there is significant negative (female lower) difference in novelty claims.

6 Conclusion

This paper examines whether team gender composition is associated with systematic differences in how economists write. Using 9,117 abstracts and an LLM-based 16-item rubric, we first show that the LLM readability score aligns with established readability measures used by Hengel 2022. We then document broad mean differences in writing style across team gender composition and show, in controlled regressions, that several of these differences remain after accounting for journal, year, and a large set of article and author characteristics.

The most robust differences are not in emotionality or generalized lack of confidence. Instead, they are concentrated in clarity and presentation: female-majority teams write more readable and direct abstracts, use less jargon, generally provide more evidentiary support, and make weaker novelty claims. Other dimensions, such as assertiveness, qualifiers, and emotional valence, show small or unstable differences.

The paper does not identify whether these differences reflect selection, training, audience expectations, or differential review pressure, and the LLM-based measures remain subject to measurement error. Even so, the results highlight writing style as a meaningful and understudied margin along which research may be evaluated, circulated, and absorbed within the discipline.

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A Rubric

Table 2: LLM Evaluation Rubric

#	Criterion	Description	Scale (1–10)
G1 Creativity & Hedging			
1	Modal Verb Strength	Balance of weak (<i>may, might</i>) vs. strong (<i>will, cannot</i>) modals	1=only weak; 10=only strong
2	Hedging Frequency & Type	Frequency of single-word qualifiers and larger stance phrases (<i>it seems that, we believe</i>)	1=continuous hedging; 10=no hedges
3	Qualifier Density	Use of modifiers (<i>to some extent, relatively</i>) that soften claims	1=extensive qualifiers; 10=none
4	Acknowledgement of Limitations	Explicit mention of caveats or boundary conditions (<i>within context, may not generalize</i>)	1=numerous limitations; 10=none or implicit only
5	Caution-Signaling Connectors	Use of <i>however, nevertheless, on the other hand</i> to signal complexity	1=frequent; 5=occasional; 10=none
G2 Assertiveness & Voice			
6	Assertiveness & Voice	Proportion of unequivocal verbs (<i>prove, confirm</i>) vs. tentative verbs (<i>suggest, explore</i>)	1=almost all tentative; 10=almost all assertive
7	Active/Passive Voice Ratio	Ratio of active (<i>we test</i>) to passive (<i>it was tested</i>) constructions	1=90% passive; 5=50/50; 10=90% active
G3 Structural Directness			
8	Sentence Length & Directness	Degree of multi-clausal, convoluted sentences vs. brief, single-clause statements	1=very long; 10=very short
9	Imperative-Form Occurrence	Presence of direct commands (<i>Apply X</i>) vs. none or only embedded suggestions	1=no imperatives; 10=dominant
G4 Authorial Stance & Novelty			
10	Pronoun Commitment	Use of first-person (<i>we, I</i>) vs. impersonal constructions (<i>the authors</i>)	1=fully impersonal; 10=fully first-person
11	Novelty-Claim Strength	Boldness of originality claims (<i>first to, novel framework</i>)	1=no novelty claim; 5=hedged; 10=grandiose
12	Jargon/Technicality Density*	Density of undefined field-specific terms vs. accessible language	1=minimal jargon; 10=dense undefined jargon
13	Emotional Valence	Presence of emotionally charged adjectives (<i>revolutionary, urgent</i>) vs. neutral descriptors	1=none; 10=frequent
G5 Support & Impact			
14	Evidence & Citation Usage	How consistently claims are backed by data, statistics, or literature citations	1=none; 10=every claim backed
15	Practical/Impact Orientation	Emphasis on real-world applications, policy, or practice implications	1=none; 5=may inform practice; 10=will transform practice
Standalone			
16	Readability	Precision and ease with which a reader can understand the author’s intended meaning	1=difficult; 10=very easy

All criteria are scored 1–10 on surface-level linguistic features only; no inference about author intent is made.

* Scale-flipped (11 – raw score) before computing the G4 composite, so that higher values consistently indicate more accessible language.

B Summary Statistics

Table 4 reports article counts by journal and decade for the full Hengel 2022 sample and the four-journal subsample used in this paper. Table 5 reports pairwise correlations among the 16 LLM-scored rubric criteria.

Table 3: LLM tonal criteria: summary statistics by gender composition

	All-Male (N = 7,951)	Mixed (N = 853)	All-Female (N = 313)	Full Sample (N = 9,117)
<i>G1: Creativity & Hedging</i>				
Modal	6.17 (2.93)	6.35 (2.85)	5.71 (3.00)	6.17 (2.93)
Hedge	8.35 (1.98)	8.49 (1.73)	8.07 (2.11)	8.35 (1.96)
Qual	8.24 (1.85)	8.32 (1.66)	8.23 (1.70)	8.25 (1.82)
AckLim	8.03 (2.56)	8.53 (2.26)	8.46 (2.24)	8.09 (2.53)
Caution	8.63 (2.24)	8.44 (2.32)	8.54 (2.29)	8.61 (2.25)
<i>G2: Assertiveness & Voice</i>				
Assert	8.18 (1.55)	8.43 (1.28)	8.04 (1.55)	8.20 (1.53)
ActPass	7.16 (2.78)	8.57 (1.78)	7.90 (2.43)	7.32 (2.73)
<i>G3: Structural Directness</i>				
Direct	5.94 (2.08)	6.71 (1.81)	6.53 (1.86)	6.03 (2.06)
Imperat	1.02 (0.22)	1.01 (0.14)	1.00 (0.00)	1.02 (0.21)
<i>G4: Authorial Stance & Novelty</i>				
Pronoun	4.08 (3.46)	6.53 (3.12)	3.98 (3.45)	4.31 (3.50)
Novelty	2.27 (2.12)	2.26 (2.20)	1.90 (1.85)	2.25 (2.12)
Jargon	7.00 (1.73)	6.48 (1.86)	6.07 (1.92)	6.92 (1.76)
Emot	1.12 (0.37)	1.11 (0.35)	1.10 (0.31)	1.12 (0.37)
<i>G5: Support & Impact</i>				
Evid	1.68 (1.34)	2.31 (1.90)	2.19 (1.73)	1.75 (1.43)
Pract	2.36 (1.73)	2.70 (1.88)	2.50 (1.80)	2.39 (1.75)
<i>Standalone</i>				
Read	6.70 (1.38)	7.42 (1.25)	7.37 (1.30)	6.79 (1.38)

Notes. Means and standard deviations (in parentheses) of LLM-scored tonal criteria, computed at the article level (AER, ECA, JPE, QJE; one observation per article after deduplication on ArticleID). Criteria scored 1–10. Gender composition is based on *Female_authorship_ratio*: All-Male (ratio=0), Mixed (0<ratio<1), All-Female (ratio=1). Short labels correspond to G-groups: G1 Creativity & Hedging; G2 Assertiveness & Voice; G3 Structural Directness; G4 Authorial Stance & Novelty; G5 Support & Impact. Readability is a standalone criterion. No significance stars shown; this table is descriptive only.

Table 4: Article count, by journal and decade

Decade	<i>AER</i>	<i>ECA</i>	<i>JPE</i>	<i>QJE</i>	Total
1950–59		120			120
1960–69		343	184		527
1970–79		660	633	1	1294
1980–89	180	648	562	401	1791
1990–99	476	443	478	409	1806
2000–09	693	519	408	413	2033
2010–15	732	382	181	251	1546
Total	2081	3115	2446	1475	9117

Notes. Included is every article published between January 1950 and December 2015 for which an English abstract was found (i) on journal websites or websites of third party digital libraries or (ii) printed in the article itself. Papers published in the May issue of *AER* (*Papers & Proceedings*) are excluded. Final row and column display total article counts by journal and decade, respectively.

C Robustness

This section estimates the key results using these alternative specifications of female authorship:

- Using a binary equal to 1 if the article’s authors are majority female
- Using a binary equal to 1 if there is 1 female coauthor
- Using only single-gender papers
- Using only solo-authored articles
- Comparing articles with a senior female coauthor to entirely male-authored papers

C.1 Robust Estimates for Table [1](#)

Table 5: Pairwise Pearson Correlations among LLM Tonal Criteria

	Modal	Hedge	Qual	AckLim	Caution	Assert	ActPass	Direct	Imperat	Pronoun	Novelty	Jargon	Emot	Evid	Pract	Read
Modal	1.00															
Hedge	0.50	1.00														
Qual	0.29	0.70	1.00													
AckLim	0.09	0.25	0.30	1.00												
Caution	0.06	0.19	0.19	0.20	1.00											
Assert	0.50	0.74	0.53	0.24	0.17	1.00										
ActPass	0.02	-0.00	0.01	0.10	-0.08	0.10	1.00									
Direct	0.11	0.20	0.21	0.25	0.09	0.20	0.31	1.00								
Imperat	0.01	0.00	-0.00	-0.06	0.01	0.01	0.00	-0.02	1.00							
Pronoun	0.04	0.06	0.04	0.07	-0.03	0.16	0.50	0.16	0.00	1.00						
Novelty	0.06	0.10	0.08	0.06	0.03	0.16	0.04	-0.03	-0.01	0.15	1.00					
Jargon	0.07	0.16	0.12	-0.10	0.09	0.16	-0.18	-0.29	0.03	-0.03	0.20	1.00				
Emot	-0.01	-0.07	-0.07	0.00	-0.06	-0.03	0.04	-0.03	-0.00	-0.00	0.08	-0.09	1.00			
Evid	0.04	-0.06	-0.08	0.06	-0.04	-0.00	0.14	0.10	-0.02	0.14	-0.02	-0.17	0.03	1.00		
Pract	-0.08	-0.11	-0.08	-0.01	-0.05	-0.08	0.11	0.03	-0.02	0.10	0.02	-0.14	0.06	0.15	1.00	
Read	0.03	0.06	0.09	0.25	-0.01	0.08	0.38	0.75	-0.05	0.22	-0.07	-0.62	0.03	0.23	0.20	1.00

Notes. Pearson correlation coefficients computed at the article level ($N = 9,117$; AER, ECA, JPE, QJE only; one observation per article). Criteria scored 1–10. Jargon/Technicality Density is shown on its raw scale (not scale-flipped). Group membership (G1–G5): G1 Creativity & Hedging; Modal, Hedge, Qual, AckLim, Caution; G2 Assertiveness & Voice: Assert, ActPass; G3 Structural Directness: Direct, Imperat; G4 Authorial Stance & Novelty: Pronoun, Novelty, Jargon, Emot; G5 Support & Impact: Evid, Pract. Readability is a standalone criterion.

Table 6: Table 1, majority female-authored

	1950–2015					1990–2015			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Flesch	1.05 *** (0.36)	1.04 *** (0.36)	1.06 *** (0.37)	1.07 *** (0.37)	1.22 *** (0.37)	1.03 *** (0.38)	1.08 *** (0.35)	1.02 *** (0.39)	1.32 *** (0.49)
Flesch-Kincaid	0.19 *** (0.09)	0.18 *** (0.09)	0.19 *** (0.09)	0.20 *** (0.09)	0.21 *** (0.09)	0.25 *** (0.10)	0.28 *** (0.09)	0.26 *** (0.10)	0.30 *** (0.11)
Gunning Fog	0.29 *** (0.11)	0.29 *** (0.11)	0.31 *** (0.11)	0.31 *** (0.11)	0.32 *** (0.11)	0.36 *** (0.12)	0.36 *** (0.11)	0.36 *** (0.13)	0.39 *** (0.13)
SMOG	0.20 *** (0.07)	0.20 *** (0.07)	0.21 *** (0.08)	0.21 *** (0.08)	0.22 *** (0.07)	0.23 *** (0.08)	0.23 *** (0.07)	0.23 *** (0.09)	0.27 *** (0.09)
Dale-Chall	0.06* (0.04)	0.06* (0.04)	0.06* (0.04)	0.07* (0.04)	0.07* (0.04)	0.07 (0.04)	0.07 (0.04)	0.07 (0.04)	0.10* (0.06)
LLM Readability	0.27 *** (0.05)	0.26 *** (0.05)	0.27 *** (0.05)	0.27 *** (0.06)	0.22 *** (0.05)	0.26 *** (0.06)	0.20 *** (0.04)	0.26 *** (0.05)	0.16 *** (0.04)
No. obs.	8,800	8,800	8,800	8,800	8,800	4,913	4,913	4,913	5,403
Editor	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blind	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal	✓	✓							
Year		✓							
Journal×Year			✓	✓	✓	✓	✓	✓	✓
N_j				✓	✓	✓	✓	✓	✓
Institution				✓	✓	✓	✓	✓	✓
Quality					✓ ¹	✓ ¹	✓ ¹	✓ ¹	✓ ¹
Native speaker					✓	✓	✓	✓	✓
JEL (primary)									
Theory/empirical									
JEL (tertiary)									✓

Notes. Estimates are identical to those in Table 1, except that female ratio has been replaced with a dummy variable equal to 1 if a weak majority (50% or more) of authors are female. (Papers with a minority—but positive—number of female authors are excluded.) ***, **, * and * statistically significant at 1%, 5% and 10%, respectively.

[H]

Table 7: Table 1, at least one female author

	1950–2015					1990–2015			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Flesch	0.51 (0.43)	0.51 (0.43)	0.52 (0.45)	0.52 (0.43)	0.63 (0.40)	0.36 (0.42)	0.34 (0.39)	0.33 (0.41)	0.38 (0.49)
Flesch-Kincaid	0.11 (0.08)	0.12 (0.08)	0.12 (0.08)	0.12 (0.08)	0.13 (0.08)	0.13 (0.08)	0.14* (0.08)	0.13 (0.08)	0.12 (0.09)
Gunning Fog	0.21* (0.10)	0.21* (0.10)	0.22** (0.11)	0.21** (0.10)	0.21** (0.10)	0.20* (0.11)	0.19* (0.10)	0.21* (0.11)	0.16 (0.11)
SMOG	0.14* (0.08)	0.14* (0.08)	0.15* (0.08)	0.14* (0.08)	0.15** (0.07)	0.13* (0.08)	0.12* (0.07)	0.13* (0.08)	0.11 (0.08)
Dale-Chall	0.05 (0.04)	0.05 (0.04)	0.06 (0.04)	0.06 (0.04)	0.07* (0.04)	0.06 (0.04)	0.06 (0.04)	0.06 (0.04)	0.06 (0.05)
LLM Readability	0.26*** (0.05)	0.25*** (0.05)	0.26*** (0.05)	0.23*** (0.05)	0.18*** (0.05)	0.20*** (0.05)	0.14*** (0.04)	0.19*** (0.05)	0.10** (0.04)
No. obs.	9.117	9.117	9.117	9.117	9.117	5.211	5.211	5.211	5.774
Editor	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blind	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal	✓	✓							
Year	✓	✓							
Journal×Year			✓	✓	✓	✓	✓	✓	✓
N_j				✓	✓	✓	✓	✓	✓
Institution				✓	✓	✓	✓	✓	✓
Quality					✓ ¹	✓ ¹	✓ ¹	✓ ¹	✓ ¹
Native speaker					✓	✓	✓	✓	✓
JEL (primary)									
Theory/empirical									
JEL (tertiary)									✓

Notes. Estimates are identical to those in Table 1, except that female ratio has been replaced with a dummy variable equal to 1 if at least one author on a paper is female. ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

Table 8: Table 1, 100% female-authored

	1950–2015				1990–2015				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Flesch	0.51 (0.64)	0.48 (0.63)	0.44 (0.64)	0.52 (0.65)	0.77 (0.66)	0.70 (0.68)	0.82 (0.66)	0.67 (0.69)	1.25 (0.96)
Flesch-Kincaid	0.13 (0.14)	0.12 (0.14)	0.11 (0.14)	0.15 (0.14)	0.17 (0.15)	0.27* (0.15)	0.31** (0.14)	0.26* (0.15)	0.34* (0.19)
Gunning Fog	0.24 (0.15)	0.24 (0.16)	0.25 (0.16)	0.30* (0.16)	0.32* (0.17)	0.46** (0.17)	0.48*** (0.16)	0.46** (0.18)	0.48** (0.21)
SMOG	0.17 (0.11)	0.17 (0.11)	0.18 (0.11)	0.20 (0.12)	0.22* (0.12)	0.30** (0.13)	0.30** (0.12)	0.29** (0.13)	0.32** (0.16)
Dale-Chall	0.06 (0.06)	0.05 (0.06)	0.05 (0.06)	0.05 (0.06)	0.07 (0.06)	0.12* (0.06)	0.13** (0.06)	0.12* (0.06)	0.17** (0.08)
LLM Readability	0.35*** (0.08)	0.34*** (0.08)	0.33*** (0.08)	0.40*** (0.08)	0.31*** (0.09)	0.40*** (0.09)	0.31*** (0.08)	0.38*** (0.08)	0.22*** (0.08)
No. obs.	8,264	8,264	8,264	8,264	8,264	4,455	4,455	4,455	4,840
Editor	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blind	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal	✓	✓							
Year		✓							
Journal×Year			✓	✓	✓	✓	✓	✓	✓
N_j				✓	✓	✓	✓	✓	✓
Institution				✓	✓	✓	✓	✓	✓
Quality					✓ ¹	✓ ¹	✓ ¹	✓ ¹	✓ ¹
Native speaker					✓	✓	✓	✓	✓
<i>JEL</i> (primary)									
Theory/empirical									
<i>JEL</i> (tertiary)								✓	✓

Notes. Estimates are identical to those in Table 1, except that female ratio has been replaced with a dummy variable equal to 1 if all authors on a paper are female. (Papers written by authors of both genders are excluded.) ***, **, * statistically significant at 1%, 5% and 10%, respectively.

[H]

Table 9: Table 1, solo-authored papers

	1950–2015				1990–2015				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Flesch	0.37 (0.74)	0.41 (0.75)	0.25 (0.74)	0.25 (0.74)	0.46 (0.75)	0.52 (0.85)	0.46 (0.81)	0.59 (0.88)	0.62 (1.75)
Flesch-Kincaid	0.13 (0.17)	0.13 (0.17)	0.11 (0.16)	0.12 (0.16)	0.09 (0.17)	0.27 (0.17)	0.27 (0.17)	0.28 (0.17)	0.10 (0.31)
Gunning Fog	0.21 (0.19)	0.22 (0.19)	0.19 (0.18)	0.21 (0.18)	0.18 (0.18)	0.40 * * (0.18)	0.36* (0.18)	0.42 * * (0.19)	0.20 (0.36)
SMOG	0.13 (0.14)	0.15 (0.14)	0.13 (0.13)	0.14 (0.13)	0.14 (0.13)	0.26* (0.14)	0.22* (0.13)	0.28* (0.14)	0.16 (0.26)
Dale-Chall	−0.01 (0.06)	−0.01 (0.07)	−0.02 (0.07)	−0.02 (0.07)	−0.01 (0.07)	0.05 (0.08)	0.06 (0.07)	0.06 (0.08)	0.10 (0.12)
LLM Readability	0.37 * ** (0.10)	0.35 * ** (0.10)	0.35 * ** (0.10)	0.35 * ** (0.10)	0.26 * ** (0.11)	0.36 * ** (0.11)	0.28 * ** (0.10)	0.35 * ** (0.10)	0.17 (0.12)
No. obs.	4,014	4,014	4,014	4,014	4,014	1,540	1,540	1,540	1,666
Editor	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blind	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal	✓								
Year		✓							
Journal×Year			✓	✓	✓	✓	✓	✓	✓
N_j				✓	✓	✓	✓	✓	✓
Institution				✓	✓	✓	✓	✓	✓
Quality					✓ ¹	✓ ¹	✓ ¹	✓ ¹	✓ ¹
Native speaker					✓	✓	✓	✓	✓
<i>JEL</i> (primary)									
Theory/empirical							✓	✓	
<i>JEL</i> (tertiary)								✓	✓

Notes. Estimates are identical to those in Table 1, except that female ratio has been replaced with a dummy variable equal to 1 if the paper is solo-authored by a woman and 0 if it is solo-authored by a man. (Co-authored papers are excluded.) ***, **, * and * statistically significant at 1%, 5% and 10%, respectively.

Table 10: Table 1, senior female author

	1950–2015				1990–2015				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Flesch	0.77 (0.65)	0.75 (0.65)	0.72 (0.66)	0.79 (0.66)	1.05 (0.66)	0.99 (0.70)	1.04 (0.65)	0.97 (0.70)	1.20 (0.91)
Flesch-Kincaid	0.17 (0.13)	0.16 (0.13)	0.16 (0.13)	0.19 (0.13)	0.21 (0.14)	0.30 ** (0.14)	0.34 ** (0.13)	0.31 ** (0.14)	0.32* (0.18)
Gunning Fog	0.31 ** (0.14)	0.31 ** (0.14)	0.32 ** (0.15)	0.36 ** (0.15)	0.38 ** (0.16)	0.51 ** (0.16)	0.51 ** (0.15)	0.51 ** (0.17)	0.45 ** (0.20)
SMOG	0.21* (0.11)	0.22* (0.11)	0.22* (0.11)	0.24 ** (0.12)	0.26 ** (0.12)	0.33 ** (0.13)	0.32 ** (0.11)	0.33 ** (0.13)	0.30* (0.15)
Dale-Chall	0.07 (0.06)	0.06 (0.06)	0.06 (0.06)	0.06 (0.06)	0.08 (0.06)	0.13* (0.07)	0.12* (0.06)	0.12* (0.07)	0.14* (0.08)
LLM Readability	0.38 ** (0.08)	0.37 ** (0.08)	0.36 ** (0.09)	0.41 ** (0.09)	0.33 ** (0.09)	0.41 ** (0.10)	0.31 ** (0.09)	0.39 ** (0.09)	0.22 ** (0.07)
No. obs.	8,396	8,396	8,396	8,396	8,396	4,581	4,581	4,581	4,991
Editor	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blind	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal	✓	✓							
Year		✓							
Journal×Year			✓	✓	✓	✓	✓	✓	✓
N_j				✓	✓	✓	✓	✓	✓
Institution				✓	✓	✓	✓	✓	✓
Quality					✓ ¹	✓ ¹	✓ ¹	✓ ¹	✓ ¹
Native speaker					✓	✓	✓	✓	✓
JEL (primary)									
Theory/empirical									
JEL (tertiary)								✓	✓

Notes. Estimates are identical to those in Table 1, except that female ratio has been replaced with the interaction between female ratio and a dummy variable equal to 1 if a female author had strictly more top-five papers as her co-authors at the time the paper was published. (Mixed-gendered papers with a senior male co-author are excluded.) ***, ** and * statistically significant at 1%, 5% and 10%, respectively.

C.2 Regression Specification for Mean differences

Table 11: Gender differences in individual LLM tonal criteria, article-level analysis (G1–G3)

	1950–2015					1990–2015			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
G1: Creativity & Hedging									
Modal Verb Strength	−0.32* (0.18)	−0.33* (0.18)	−0.34* (0.19)	−0.32* (0.19)	−0.34* (0.19)	−0.29 (0.22)	−0.26 (0.21)	−0.30 (0.22)	−0.31 (0.21)
Hedging Frequency & Type	−0.21* (0.11)	−0.21* (0.11)	−0.23** (0.11)	−0.20* (0.11)	−0.17 (0.11)	−0.19 (0.13)	−0.14 (0.13)	−0.19 (0.13)	−0.19 (0.12)
Qualifier Density	−0.02 (0.10)	−0.01 (0.10)	−0.03 (0.10)	0.00 (0.09)	0.01 (0.10)	0.06 (0.10)	0.07 (0.10)	0.06 (0.10)	−0.06 (0.12)
Acknowledgement of Limitations	0.23* (0.13)	0.24** (0.12)	0.23* (0.12)	0.24** (0.11)	0.15 (0.12)	0.18 (0.13)	0.14 (0.13)	0.17 (0.13)	0.10 (0.14)
Caution-Signaling Connectors	0.06 (0.13)	0.08 (0.13)	0.09 (0.13)	0.08 (0.14)	0.08 (0.14)	0.04 (0.17)	0.07 (0.18)	0.04 (0.17)	−0.03 (0.18)
G2: Assertiveness & Voice									
Assertiveness & Voice	−0.08 (0.08)	−0.08 (0.08)	−0.10 (0.08)	−0.06 (0.07)	−0.05 (0.08)	−0.07 (0.09)	−0.04 (0.09)	−0.07 (0.09)	−0.07 (0.09)
Active/Passive Voice Ratio	0.07 (0.10)	0.06 (0.10)	0.07 (0.10)	0.16 (0.10)	0.15 (0.10)	0.18* (0.11)	0.19* (0.11)	0.20* (0.11)	0.23** (0.11)
G3: Structural Directness									
Sentence Length & Directness	0.27** (0.10)	0.27** (0.11)	0.27** (0.11)	0.31*** (0.10)	0.25** (0.10)	0.37*** (0.10)	0.36*** (0.10)	0.37*** (0.10)	0.31*** (0.10)
Imperative-Form Occurrence	−0.02*** (0.01)	−0.02*** (0.01)	−0.02*** (0.01)	−0.02*** (0.01)	−0.02*** (0.01)	−0.01** (0.01)	−0.01* (0.01)	−0.01** (0.01)	−0.01* (0.00)
No. obs.	9,117	9,117	9,117	9,117	9,117	5,211	5,211	5,211	5,774
Editor	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blind	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal×Year			✓	✓	✓	✓	✓	✓	✓
N_j				✓	✓	✓	✓	✓	✓
Institution				✓	✓	✓	✓	✓	✓
Quality				✓	✓	✓	✓	✓	✓
Native speaker					✓	✓	✓	✓	✓
<i>JEL</i> (primary)					✓	✓	✓	✓	✓
Theory/empirical									
<i>JEL</i> (tertiary)									

Notes. Figures represent coefficients on female ratio from 81 separate OLS regressions of individual LLM tonal criteria on the ratio of female authors. Each row is a separate criterion. Criteria belong to groups G1 (Creativity & Hedging), G2 (Assertiveness & Voice), and G3 (Structural Directness). ***, **, * and * statistically significant at 1%, 5% and 10%, respectively.

Table 12: Gender differences in individual LLM tonal criteria, article-level analysis (G4–G5 and Readability)

	1950–2015			1990–2015					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
G4: Authorial Stance & Novelty									
Pronoun Commitment	−0.09 (0.18)	−0.10 (0.18)	−0.10 (0.18)	0.39 ** (0.19)	0.32 (0.20)	0.53 ** (0.24)	0.54 ** (0.23)	0.53 ** (0.24)	0.61 *** (0.20)
Novelty-Claim Strength	−0.32 *** (0.10)	−0.32 *** (0.11)	−0.32 *** (0.11)	−0.31 *** (0.11)	−0.30 *** (0.11)	−0.36 *** (0.14)	−0.30 *** (0.14)	−0.36 *** (0.14)	−0.30 *** (0.13)
Jargon/Technicality Density [†]	0.52 ***	0.52 ***	0.53 ***	0.55 ***	0.48 ***	0.54 ***	0.37 ***	0.52 ***	0.41 ***
Emotional Valence	(0.11)	(0.11)	(0.11)	(0.12)	(0.12)	(0.15)	(0.11)	(0.14)	(0.12)
	−0.02 (0.02)	−0.02 (0.02)	−0.03 (0.02)	−0.03 (0.02)	−0.03 (0.02)	−0.03 (0.02)	−0.03 (0.02)	−0.03 (0.02)	−0.03 (0.02)
G5: Support & Impact									
Evidence & Citation Usage	0.44 *** (0.09)	0.44 *** (0.09)	0.44 *** (0.09)	0.46 *** (0.09)	0.36 *** (0.09)	0.37 *** (0.11)	0.26 *** (0.10)	0.33 *** (0.11)	0.16 (0.10)
Practical/Impact Orientation	0.06 (0.09)	0.06 (0.09)	0.07 (0.10)	0.09 (0.09)	0.05 (0.10)	−0.01 (0.11)	−0.11 (0.10)	−0.01 (0.11)	−0.15 (0.09)
Standalone									
Readability	0.36 *** (0.07)	0.35 *** (0.07)	0.35 *** (0.07)	0.40 *** (0.08)	0.32 *** (0.08)	0.39 *** (0.08)	0.31 *** (0.06)	0.38 *** (0.07)	0.27 *** (0.06)
No. obs.	9.117	9.117	9.117	9.117	9.117	5.211	5.211	5.211	5.774
Editor	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blind	✓	✓	✓	✓	✓	✓	✓	✓	✓
Journal	✓	✓							
Year		✓							
Journal×Year			✓	✓	✓	✓	✓	✓	✓
N_j				✓	✓	✓	✓	✓	✓
Institution				✓	✓	✓	✓	✓	✓
Quality					✓ ¹	✓ ¹	✓ ¹	✓ ¹	✓ ¹
Native speaker					✓	✓	✓	✓	✓
<i>JEL</i> (primary)									
Theory/empirical								✓	
<i>JEL</i> (tertiary)									✓

Notes. Figures represent coefficients on female ratio from 63 separate OLS regressions of individual LLM tonal criteria on the ratio of female authors. Each row is a separate criterion. Criteria belong to groups G4 (Authorial Stance & Novelty), G5 (Support & Impact), and the standalone Readability criterion. [†] Jargon/Technicality Density is scale-flipped (11 – raw score) so that higher values indicate more accessible language. See companion table for G1–G3. ***, **, * and * statistically significant at 1%, 5% and 10%, respectively.